Amir Reza Vazifeh

amir.vazifeh@princeton.edu | conavaz@partner.waters.com | amirrezavazifeh.github.io

RESEARCH INTERESTS

Computational Imaging & Optics | Computer Vision | Inverse Problems | AI in Medicine & Healthcare

EDUCATION

Princeton University 2023 – 2028
Ph.D., Electrical & Computer Engineering, Minor in Computational Finance *GPA 3.9/4.0*

- **Major Coursework:** Machine Learning & Pattern Recognition | Deep Learning Networks | Computer Vision | Imaging the Invisible | Biomedical Imaging | Nonlinear Optics | Digital Signal Processing | Advanced Wireless Systems
- **Minor Coursework:** Statistical Machine Learning | Computational Finance in C++

Sharif University of Technology 2018 – 2022
B.S., Electrical Engineering, Minor in Physics — Summa Cum Laude *GPA 18.83/20*

RESEARCH EXPERIENCE

Princeton University 2024 – Present
Graduate Researcher — Advisor: Prof. Jason Fleischer; Collaborator: Prof. Felix Heide *Princeton, NJ*

- Developed an unsupervised, video-based image reconstruction algorithm for fiber bundle imaging systems using implicit neural representations
- Created a framework for personalized, unsupervised arrhythmia detection using manifold learning on multi-lead electrocardiogram signals, an interactive software demo is available online
- Developed a snapshot hyperspectral imaging method with randomized aperture codes and neural fields

Institute of Science and Technology Austria Summer 2023
Research Intern (among top 3% of admitted applicants) — Advisor: Prof. Sandra Siegert *Vienna, Austria*

- Implemented a 3D instance segmentation model for microglia cells

École Polytechnique Fédérale de Lausanne (EPFL) Summer 2022
Research Intern (among top 2% of admitted applicants) — Advisor: Prof. Mahsa Shoaran *Geneva, Switzerland*

- Analyzed phase-amplitude cross-frequency coupling in EEG signals

Sharif University of Technology 2020 – 2022
Undergraduate Researcher — Advisors: Profs. Mohammad Fakharzadeh & Daryoosh Vashaee *Tehran, Iran*

- Designed an ultralow-power wearable ECG patch for long-term cardiac monitoring, paired with a cloud platform for AI-based denoising and arrhythmia detection
- Reviewed energy harvesting materials and techniques for self-powered wearable medical devices

INDUSTRY EXPERIENCE

Waters Corporation (Industry-PhD Partnership) 2024 – Present
Computer Vision and Machine Learning Researcher *Remote, US*

- Built **ViabiLens**, AI-assisted software for label-free cell counting and viability prediction from brightfield images; an interactive software demo is available online
- Driving Waters' next-generation imaging platform; recognized by senior leadership up to SVP level
- Co-inventor on a U.S. provisional patent for label-free cell viability assessment

PATENTS

Cell Viability Assessment — U.S. Provisional Patent Application, Waters Corporation Filed June 2026
C. Zeigler, **A. R. Vazifeh**, S. Meyyappan, R. Jeske, J. W. Fleischer

PUBLICATIONS

- **Under review:** Label-Free Cell Counting and Viability Prediction Using Brightfield Imaging and Deep Learning, **A. R. Vazifeh**, C. Zeigler, S. Meyyappan, R. Jeske, J. W. Fleischer. *arXiv*, 2026
- Neural Fields for Single-Shot Image Reconstruction in Fiber Bundle Imaging Systems, **A. R. Vazifeh**, J. W. Fleischer. *Frontiers in Optics*, 2026
- AI-Enabled Label-Free Cell Viability Assessment from Brightfield Imaging, **A. R. Vazifeh**, C. Zeigler, S. Meyyappan, R. Jeske, J. W. Fleischer. *Frontiers in Optics*, 2026

- Manifold Alignment for Label-Free Cell Phenotyping in Multimodal Microscopy, **A. R. Vazifeh**, J. W. Fleischer. *Optica Biophotonics Congress (Microscopy, Histopathology and Analytics)*, 2026
- Seeing Through Fibers: Unsupervised Image Reconstruction in Fiber Bundle Imaging Systems, **A. R. Vazifeh**, C. Wang, A. Joshi, I. Chugunov, J. Sun, J. Yeom, J. W. Fleischer, J. S. Pulido, F. Heide. *Optics Express*, 2026
- Manifold Learning for Personalized and Label-Free Detection of Cardiac Arrhythmias, **A. R. Vazifeh**, J. W. Fleischer. *Informatics in Medicine Unlocked*, 2026
- An Open-Source Retrospective Analysis of Hypertrophic and Dilated Cardiomyopathy Using Machine Learning and Electrocardiogram Data, A. Altintepe, A. Rustemli, **A. R. Vazifeh**, J. W. Fleischer. *Diagnostics*, 2026
- Snapshot Hyperspectral Imaging via Compressive Sensing and Implicit Neural Representation, J. Boondicharern, **A. R. Vazifeh**, J. W. Fleischer. *Optica Imaging Congress (Computational Optical Sensing and Imaging)*, 2025
- Advancing Personalized Healthcare and Entertainment: Progress in Energy Harvesting Materials and Techniques of Self-Powered Wearable Devices, P. Bhatnagar, S. Hooshmand Zaferani, N. Rafiefard, B. Baraeinejad, **A. R. Vazifeh**, R. Mohammadpour, R. Ghomashchi, H. Dillersberger, D. Tham, D. Vashaee. *Progress in Materials Science*, 2023
- Design and Implementation of an Ultralow-Power ECG Patch and Smart Cloud-Based Platform, B. Baraeinejad, M. Fallah Shayan, **A. R. Vazifeh**, D. Rashidi, M. Saberi Hamedani, H. Tavolinejad, P. Gorji, P. Razmara, K. Vaziri, D. Vashaee, M. Fakharzadeh. *IEEE Transactions on Instrumentation and Measurement*, 2022

HONORS & AWARDS

- ECE Graduate Student Award for Excellence in Service, Princeton University 2026
- Ph.D. Fellowship in Natural Sciences and Engineering, Princeton University 2023
- Academic Achievement Award, Sharif University of Technology 2020
- Ranked 12th among 134,000+ applicants, Iranian National University Entrance Exam 2018

TEACHING & MENTORSHIP

- Assistant in Instruction, ECE 452: Biomedical Imaging, Princeton University Spring 2025, 2026
- Assistant in Instruction, ECE 535: Machine Learning & Pattern Recognition, Princeton University Fall 2024
- Research Mentor for high school students, Princeton Laboratory Learning Program Summer 2025

SKILLS

- **Programming & ML:** Python | C/C++ | HTML/CSS | Git | LaTeX
- **Cloud & Data:** AWS (S3, EC2), large-scale data analysis, distributed/parallel computing
- **Hardware:** Circuit & board design (Altium, HSPICE, Proteus), Verilog, MIPS Assembly

PEER REVIEW

- *IEEE Journal of Biomedical and Health Informatics*
- *The Journal of Supercomputing* (Springer Nature)